



Machine Learning Object-Based Image Analysis for Urban Land Use Classification Near California Light Rail

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Overview

- Gathered coordinate data for all California light rail stations across 5 systems (sacRT, SFMTA, VTA, LAM, MTS) via GTFS transit feeds and webscraping/XML parsing
- Reprojected coordinates into appropriate NAD83 UTM State Plane Projected Coordinate Systems (PCS)
- Created 400m station buffers
- Extracted high resolution (0.6m) station rasters from buffer extents via Microsoft's Planetary Computer, utilized National Agriculture Imagery Program (NAIP), includes system for raster mosaic for split tiles
- Segmented rasters with graph-based methods and segment post-processing into meaningful objects for Object-Based Image Analysis (OBIA)
- Extracted features from segments via remote sensing techniques (NDVI, NDWI, GLCM texture, etc)
- Classified objects by land use (LULC) using Random Forest, assessed land use mix of stations/systems



Project Objectives



- Understand how land is used around California transit stops
- Understand systemic differences in land use
- Better understand the differences in infrastructure types
- Explore ML Land Use Land Cover (LULC) classification via Object-Based Image Analysis (OBIA)

Concepts Used

Data/Image Processing and Cleaning

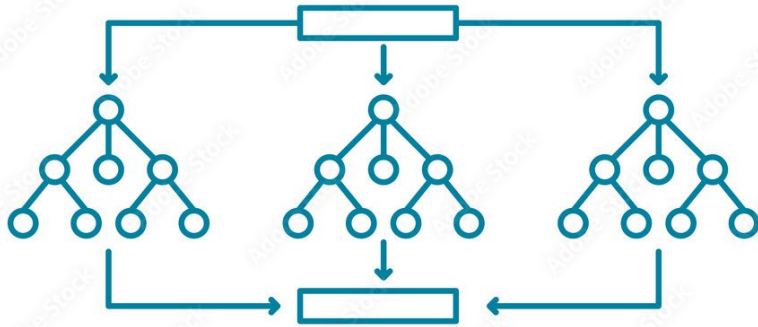
- Rasterio/Shapely/Ski-Image
- Geopandas
- Microsoft Planetary Computer
- Webscraping/XML Parsing
 - Requests, re, lxml
- 'Fuzzy' string matching
- Raster reprojection
- Coordinate Buffering
- Imagery Tile Mosaics
- White balance/Histogram Equalization, contrast enhancement
- Band normalization



A Planetary
Computer for a
Sustainable Future

The image shows a black rectangular box containing white text. To the right of the text is a graphic of a bright sun or star rising over the horizon of a planet, with blue and white light rays emanating from the point of contact. The planet's surface is visible as a thin blue and white arc.

Machine Learning



- Utilized random forest classification for final model
- Experimented with U-Net++ and Mask R-CNN Neural Networks for deep learning classification
- Tested several ML segmentation algorithms - Watershed, MeanShift, etc
- Concurrent.futures

GIS

- QGIS
- Coordinate/Raster Reprojection
- Buffers/Contains/Intersects
- Tile Mosaicking
- Index/GLCM Textural features
- National Agriculture Imagery Program (NAIP) Satellite Imagery
- Remote Sensing





Framework and Walkthrough

Defining Project Scope



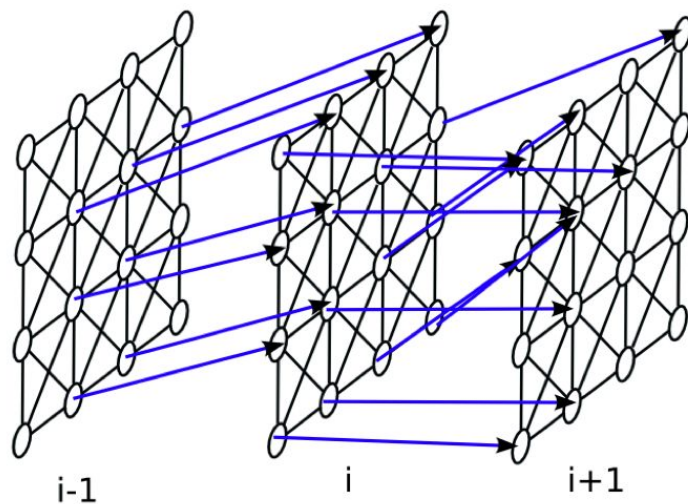
- Chose California Light Rail, compared 5 systems in state that have light rail transit:
 - Sacramento Regional Transit (sacRT) - Sacramento/Sacramento County
 - San Francisco Muni Metro (SFMTA) - San Francisco
 - Valley Transportation Authority (VTA) - San Jose/Santa Clara County
 - Los Angeles Metro (LAM) - Los Angeles/Los Angeles County
 - San Diego Trolley (MTS) - San Diego/San Diego County
- Chose 5 Categories of Interest:
 - Buildings, Parks/Open Space, Undeveloped Land, Transportation, Water
- Systems exist in California context, utilize same technology and cover similar geographic scales

Data Acquisition/Preprocessing

- Coordinate Data Acquisition:
 - Created geodataframes for system, stop, coordinates (lat/long GCS, degrees, and PCS, meters)
 - For SFMTA and MTS, webscraped data to retrieve relevant stops
 - Buffered coordinates to retrieve polygon of 400m 'walkshed' for each station
- Image Data Acquisition:
 - Utilized NAIP imagery (0.6m resolution), queried Microsoft's Planetary Computer to retrieve all tiles that encompassed each system, window read relevant subtile for buffers contained by single NAIP tile, mosaicked multiple tiles for edge case of station being intersected
- Image processing:
 - Normalized image bands (0-1 scale) and reorganized channels so ski-image could be used
 - Improved white balance and increased pixel contrast via ski-image, equalized RGB histograms

Image Segmentation

- Segmented satellite images into semantically significant objects for Object Based Image Analysis (OBIA)
 - Utilized instance segmentation with rule-based methods (as opposed to deep learning)
- Graph based algorithm chosen - Felzenszwalb Segmentation Algorithm
- Segmentation post-processing
 - Smoothing and polygon simplification
 - Small polygon merging based on predefined thresholds



Feature Extraction, Training Labeling, and Object Classification

- For each segment in each image, extracted:
 - Statistical Features: min, mean, max, and std reflectance
 - Spectral Features: mean and standard deviation of intensity per band (R, G, B, NIR)
 - Index Features: NDVI (Normalized Difference Vegetation Index), NDWI (ND Water Index), MSAVI2 (Second Modified Soil Adjusted Vegetation Index), SAVI (Soil Adj. Veg. Idx.)
 - Textural Features: GLCM (contrast, entropy, homogeneity, correlation per band)
 - Shape Features: Solidity, First 3 Hu Moments
- Manually labeled 2,000 segments for 2 stations per system via QGIS
- Classified objects utilizing Random Forest Classification, GridSearchCV used for hyperparameter tuning/cross validation
 - Best Parameters
 - max depth = 30
 - max features = 'sqrt'
 - min samples leaf = 2
 - min samples split = 2
 - number estimators = 200

Results and Analysis

82% CV Model Accuracy!

- Precision of 0.91 for parks/open space
- Lowest accuracy for water - f1-score of 0.67
- Accuracy improved after simplifying classes

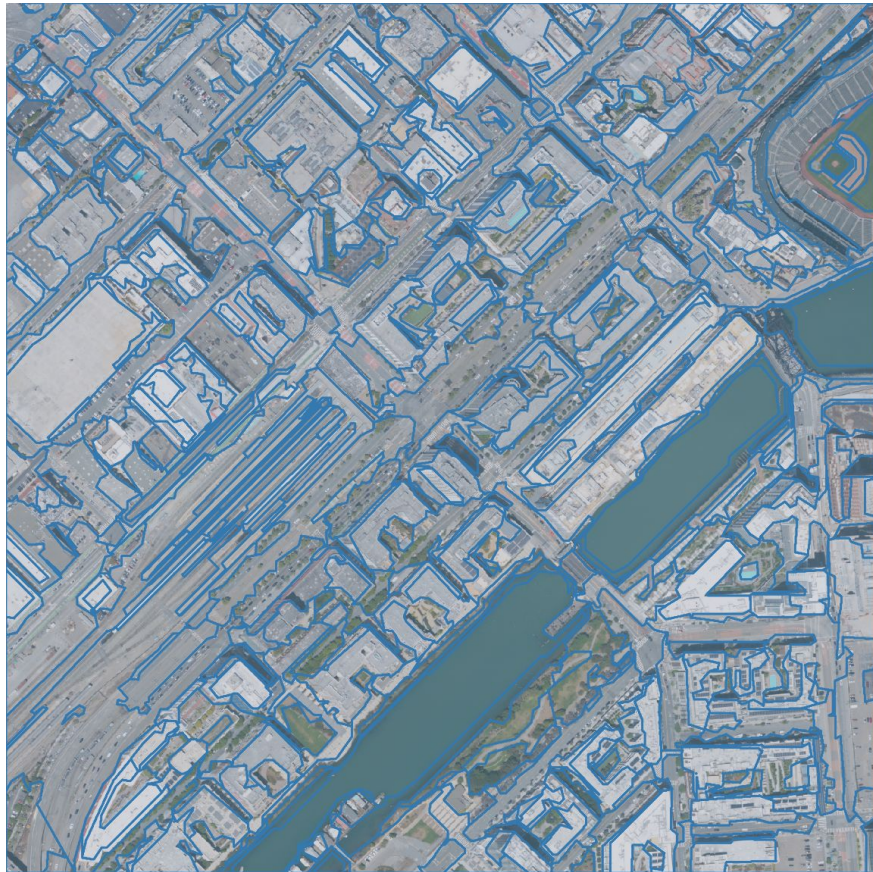
Analysis

- Model evaluation metrics based on correct segment classifications, does not consider relative size of segment
 - Small polygon sliver artifacts created from segmentation imperfections
 - When considering area of segment, model had much higher accuracy for 'normal' sized segments
- Model had trouble distinguishing between building shadows and water
- Overall strong performance, high practical accuracy
- Model excels at detecting buildings, does not perform well with more granular classification





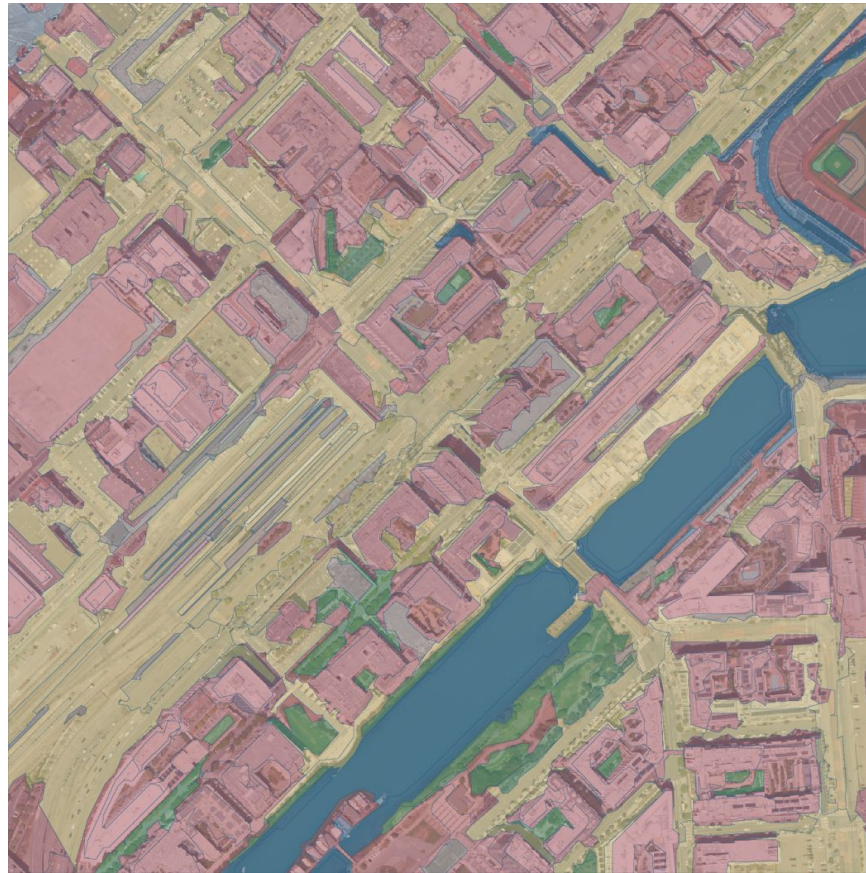
Original NAIP Image - 4th and King, SFMTA



Segmented Image - 4th and King, SFMTA

Legend

	Buildings
	Transportation
	Parks/Open Space
	Water
	Undeveloped



Land Use Proportions

Buildings	46.71%
Transportation	37.42%
Parks/Open Space	3.90%
Water	8.49%
Undeveloped	3.48%

Classified Image - 4th and King, SFMTA



Original NAIP Image - Archives Plaza Station, sacRT



Segmented Image - Archives Plaza Station, sacRT

Legend

	Buildings
	Transportation
	Parks/Open Space
	Water
	Undeveloped



Land Use Proportions

Buildings	48.20%
Transportation	19.73%
Parks/Open Space	31.08%
Water	0%
Undeveloped	0.99%

Classified Image - Archives Plaza Station, sacRT



Original NAIP Image - Moffett Park Station, VTA



Segmented Image - Moffett Park Station, VTA

Legend

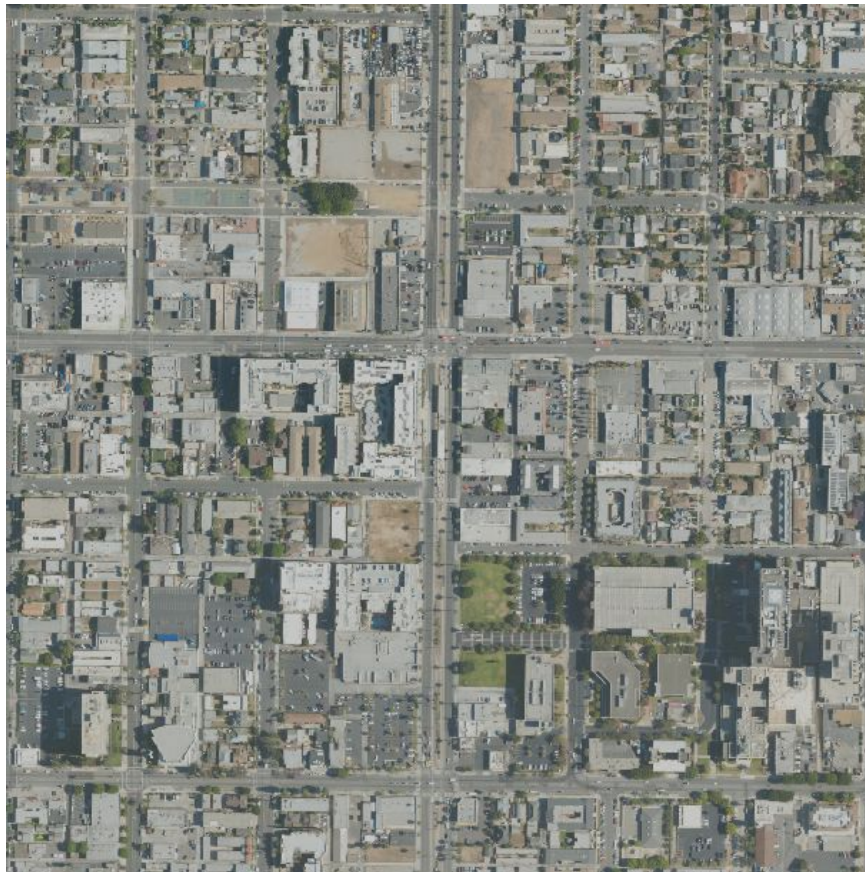
	Buildings
	Transportation
	Parks/Open Space
	Water
	Undeveloped



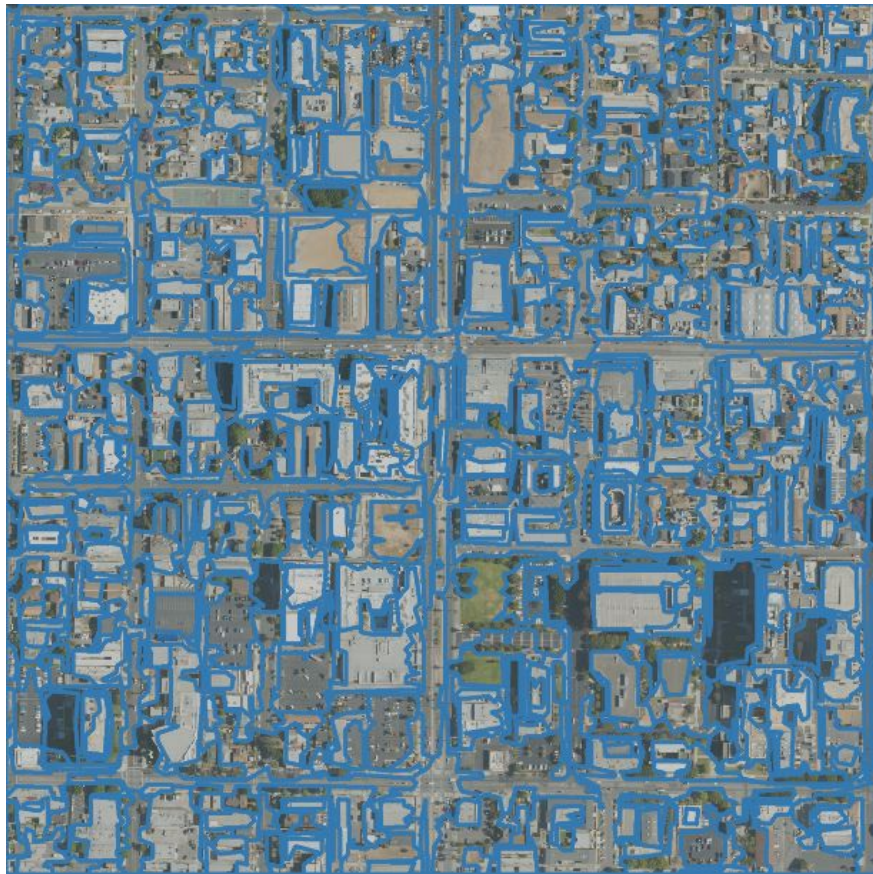
Land Use Proportions

Buildings	35.06%
Transportation	36.88%
Parks/Open Space	18.12%
Water	0%
Undeveloped	9.95%

Classified Image - Moffett Park Station, VTA



Original NAIP Image - Anaheim Street Station, LA Metro



Segmented Image - Anaheim Street Station, LA Metro

Legend

	Buildings
	Transportation
	Parks/Open Space
	Water
	Undeveloped



Land Use Proportions

Buildings	68.22%
Transportation	26.74%
Parks/Open Space	2.75%
Water	0.29%
Undeveloped	2.01%

Classified Image - Anaheim Street Station, LA Metro

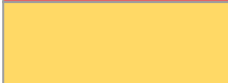


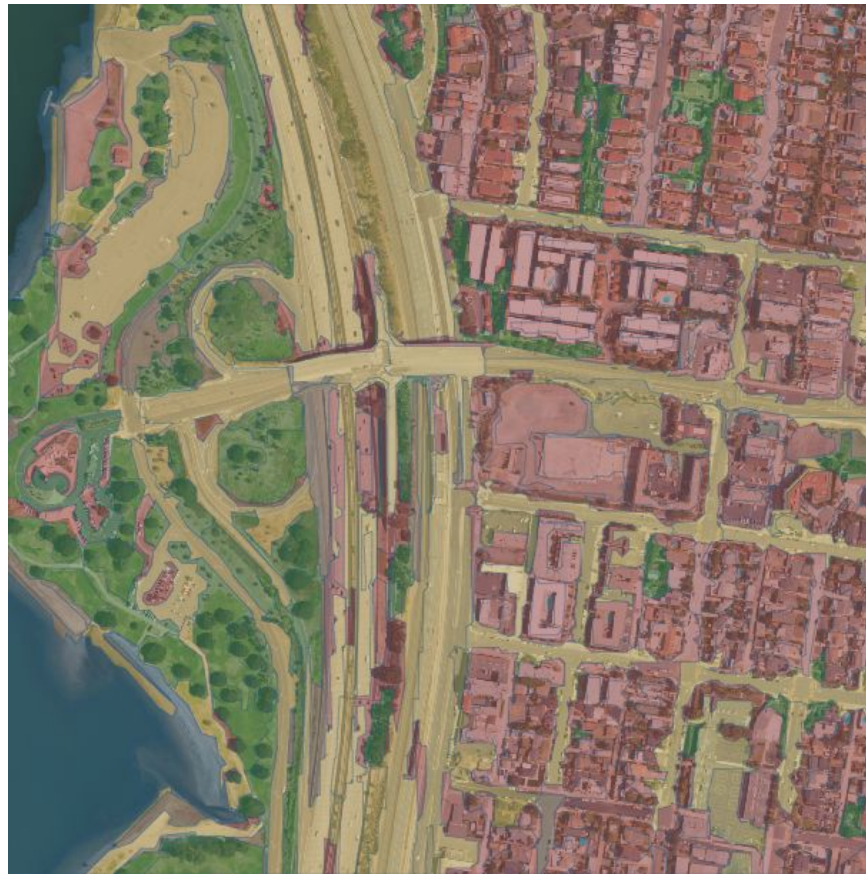
Original NAIP Image - Clairemont Drive Station, MTS



Segmented Image - Clairemont Drive Station, MTS

Legend

	Buildings
	Transportation
	Parks/Open Space
	Water
	Undeveloped



Land Use Proportions

Buildings	42.49%
Transportation	31.73%
Parks/Open Space	16.47%
Water	5.95%
Undeveloped	3.36%

Classified Image - Clairemont Drive Station, MTS

Transit System Comparative Analysis



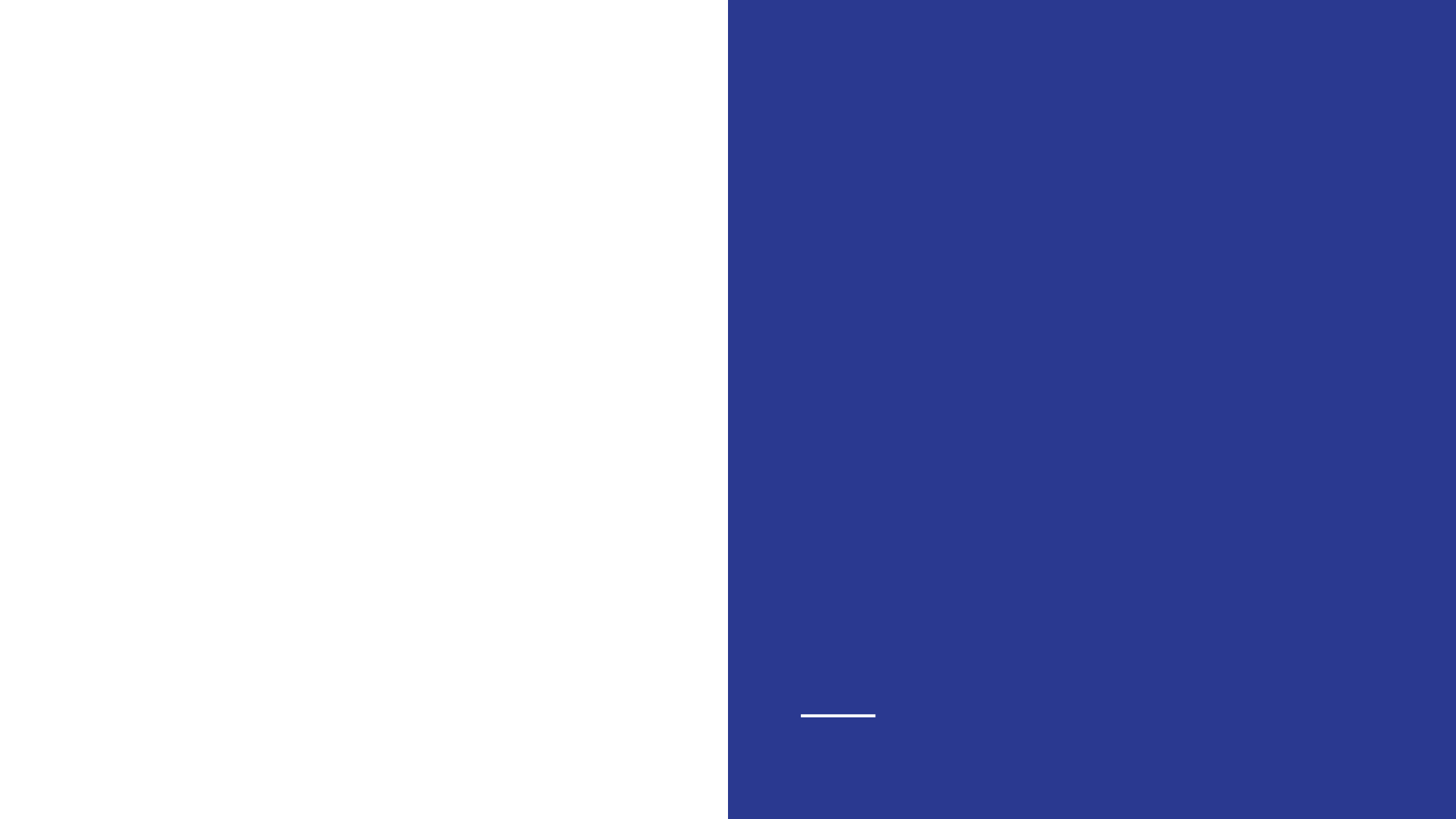
System	Urban%	Transp.%	Undev.%	Parks%	Water%
SFMTA	64.4%	25.6%	1.9%	7%	1%
sacRT	51.8%	27.5%	5.3%	14.7%	0.7%
VTA	54.2%	30.3%	4.1%	11%	0.4%
LAM	60.6%	30.2%	3%	5.2%	0.9%
MTS	50.8%	31.7%	4.8%	11.3%	1.5%

Systems vary somewhat significantly by average land use!

Future Work and Conclusions



- Geospatial Machine Learning can be leveraged to make large scale comparisons between system design and surrounding land use
 - Further improve segmentation accuracy to improve model performance
 - Introduce Shadow removal techniques for better accuracy
 - Transit Oriented Development (TOD) can remedy poor land use
 - Compare specific stations contextualized by ridership
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Thank you!

